

HORMONE RECEPTORS AND SIGNAL TRANSDUCTION

Hormone molecules exert their effects by producing biochemical changes in target cells. Hormones first bind to specific receptor molecules. It initiates changes in target cells. Some hormones bind to intracellular receptors and others to cell-surface receptors. Three types of receptor molecules are important in mediating hormone actions. These are:

- a) Intracellular receptors
 - b) G protein-coupled membrane receptors
 - c) Enzyme-linked membrane receptors
- (a) **Intra-cellular receptors**

(i) **Hormone genomic effect**

Lipid-soluble hormones are like steroids and iodothyronines. They are lipid soluble. Hence they bind to **intracellular receptors**. These hormones enter target cells by diffusing through lipid bilayer of cell membrane or carried bound to a lipoprotein molecule. Their receptors are located in the cytoplasm or in the nucleus. A hormone molecule binds to receptor molecule. As a result, it forms a **hormone-receptor complex**. This complex acts as a **transcription factor**. It interacts with target DNA of cell to alter gene expression. The hormone turns the processes of transcription and translation on or off. Thus hormone directly influences the synthesis of proteins by the target cell. New proteins can be enzymes or structural proteins. These enzyme or proteins carry out target cell's physiological response. The production of new proteins requires time. Therefore, there is a delay (ranging from many minutes to hours) between hormone's binding to intracellular receptor and the target cell's response. Once the proteins are synthesized, the response lasts until they are degraded.

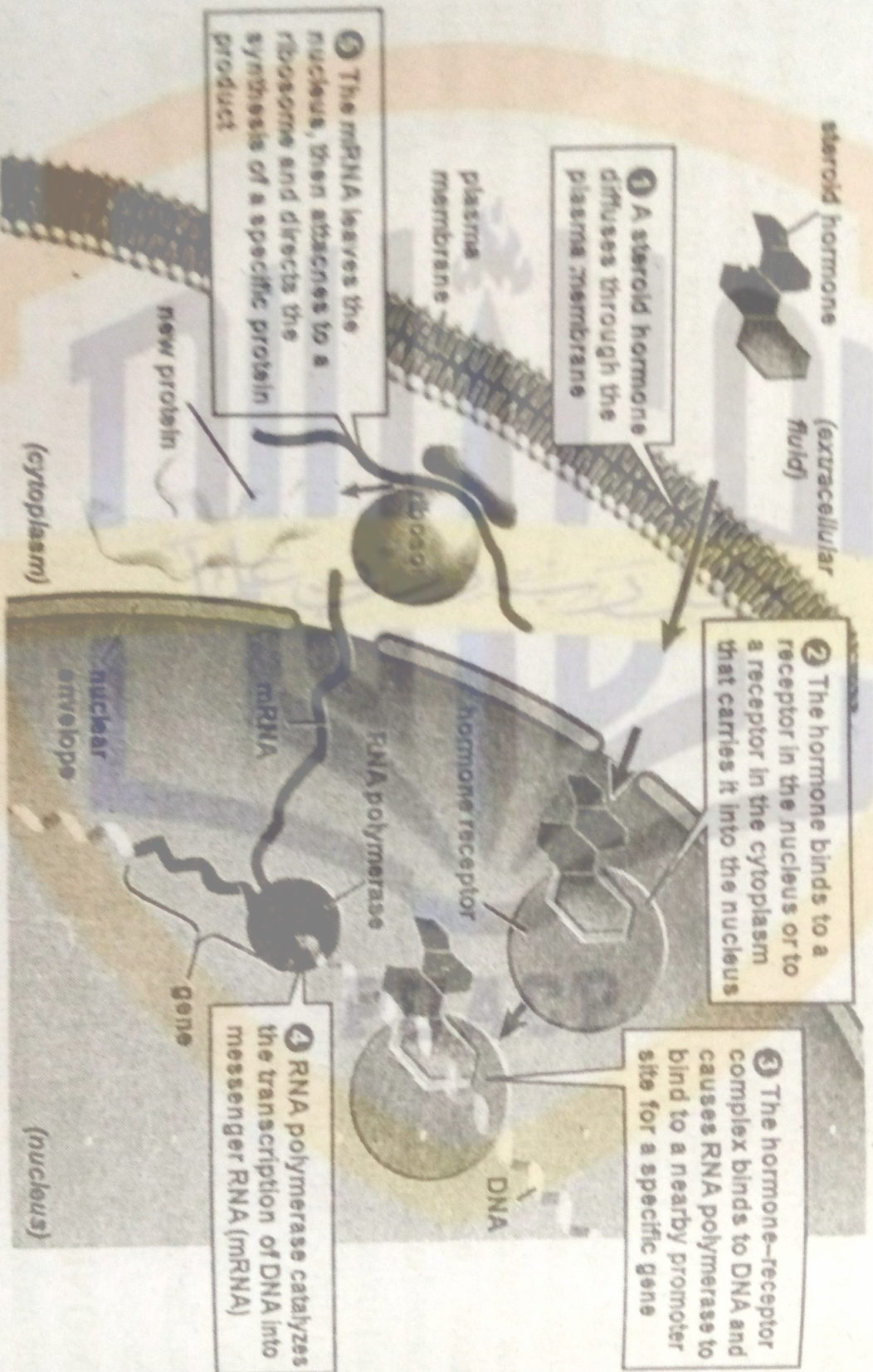


Fig: Steroid hormone action

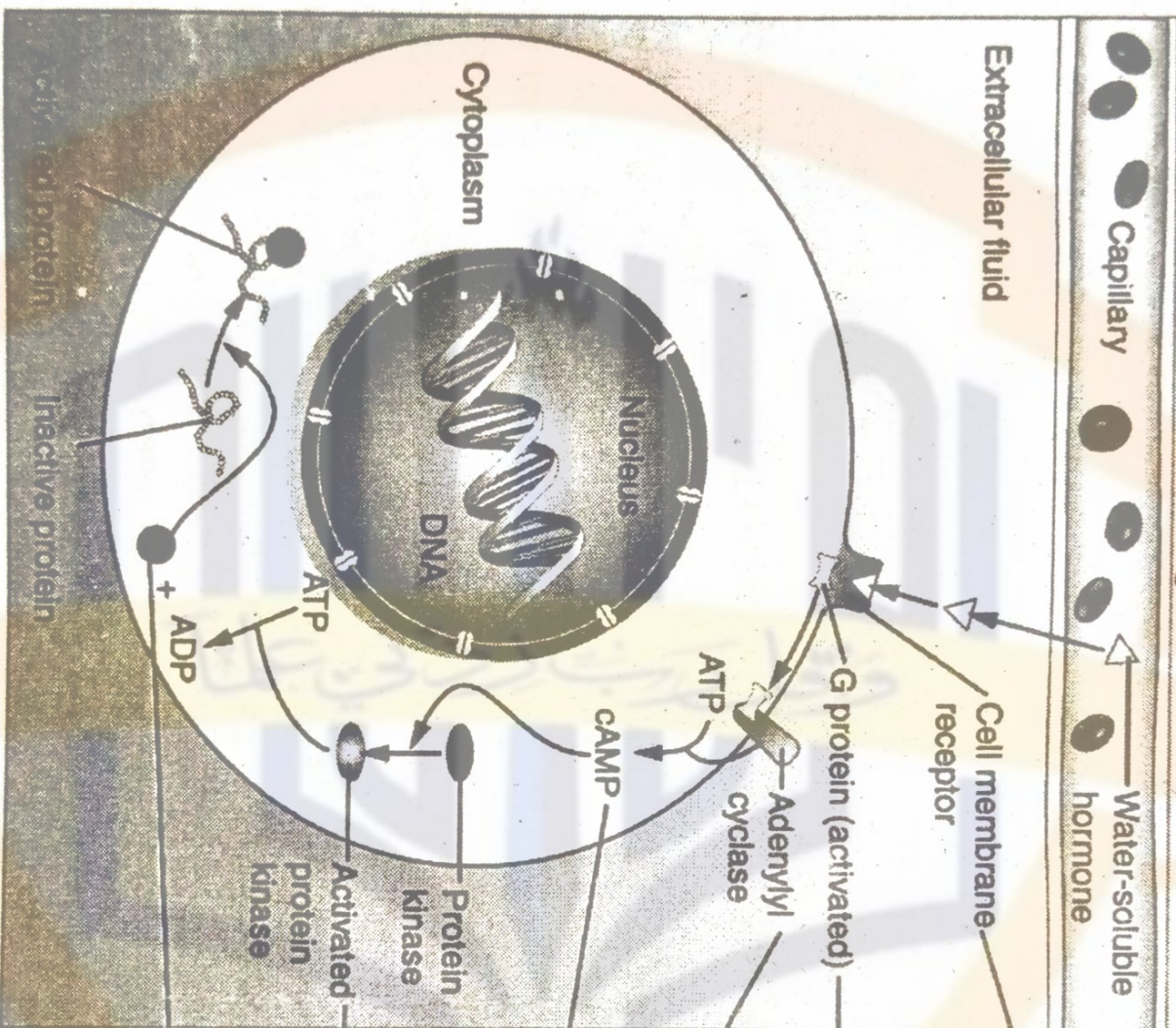


Fig: G protein-coupled membrane receptors

(ii) Hormone non-genomic effect

Certain target cells of some steroid hormones like aldosterone, estrogen, and certain glucocorticoids, express cell-surface receptors for hormones. The steroid hormones bind to receptors on the cell membrane. They do not exert their effects by genomic means. But they function like water-soluble hormones. They show rapid effects called **non-genomic effects**.

(b) G protein-coupled membrane receptors

The water-soluble peptide and catecholamine hormones do not diffuse into cells. Their receptor molecules are located in the cell membrane. They have external hormone-binding sites. These receptors mediate hormone actions by altering membrane permeability or by activating an intracellular second-messenger system by a G protein. Second messengers are like cAMP, cGMP, Ca^{2+} ions and phosphoinositides. They play an important role in transducing hormonal signal into a response of target cell.

(c) Enzyme-linked membrane receptors

A few peptide hormones like insulin and growth hormone bind to enzyme-linked membrane receptors. Peptide hormones exert their effects by changing activities of existing proteins. Some may also alter gene activities. Target cells can change their physiology quickly by using preexisting proteins. Hence their responses to these hormones can be measured within minutes.

SYNTHESIS, STORAGE AND RELEASE OF HORMONES

Two chemical classes of hormones: peptides and steroids have two different mechanisms of synthesis, storage, and secretion.

(a) Synthesis of Steroid Hormones

We use insulin as our primary example of a peptide hormone. Insulin is the vertebrate hormone that lowers the concentration of blood glucose. It stimulates the cells to take up glucose from the blood. It also promotes the incorporation of food molecules into storage compounds. Insulin is synthesized in endocrine cells in the pancreas. Pancreatic endocrine cells are clustered in the islets of Langerhans. These islets are embedded in the pancreatic exocrine tissue. The islet cells that produce and secrete insulin are called β or (beta) cells. Insulin follows pattern of typical synthesis of most secreted peptides. In this pattern a large precursor molecule, a **preprohormone** is modified by posttranslational processing. It is changed into a **prohormone** and finally a mature hormone.

Peptide hormones are synthesized at ribosomes. They are stored in vesicles. Then they are secreted on demand. The insulin molecule consists of two peptide chains called A and B. These chains are connected by disulfide bonds.

Process of Synthesis of Insulin**1. Formation of preproinsulin**

a) Transcription: Amino acid sequences of the chains are encoded in DNA of the cell nucleus. The DNA nucleotide sequence of the insulin gene is transcribed into nucleotide sequence of messenger RNA (mRNA). The mRNA leaves the nucleus and interacts with ribosomes of rough (granular) endoplasmic reticulum (ER).

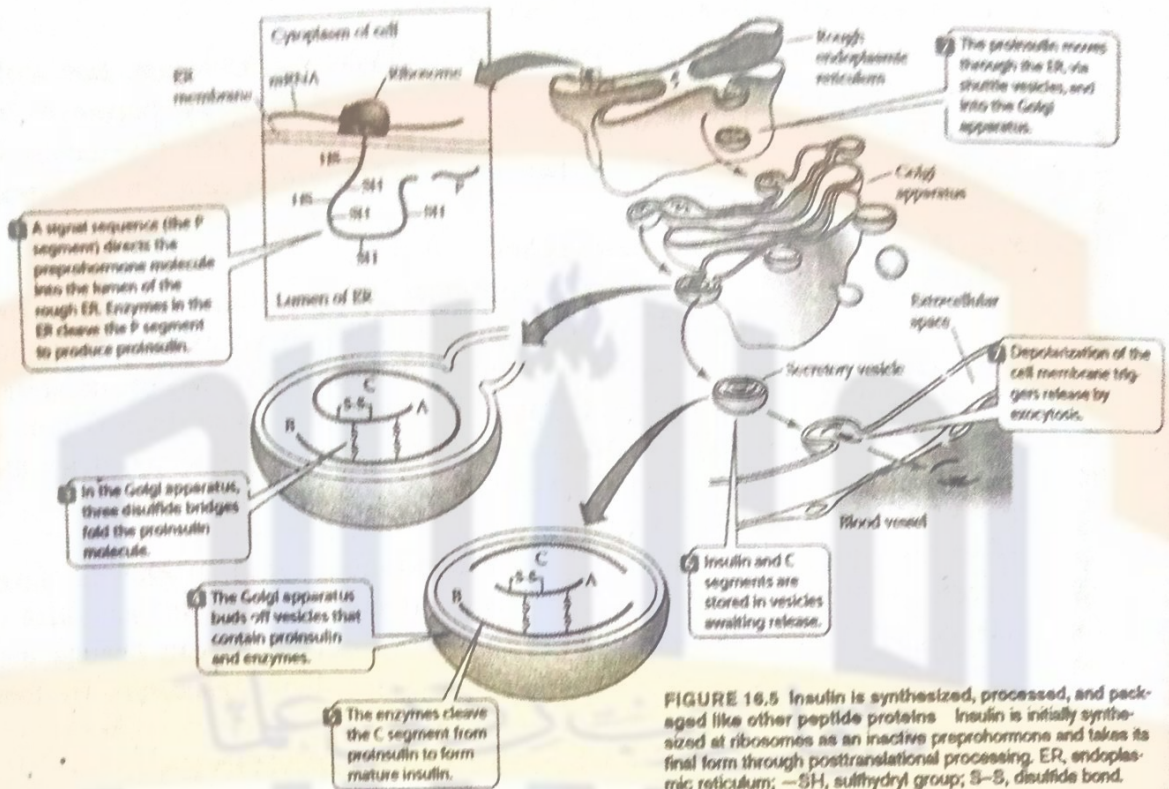


Fig: Insulin is synthesized, processed, and packaged like other peptide proteins

- b) **Translation:** The nucleotide sequence of mRNA is then translated into amino acid sequence of the polypeptide called **preproinsulin**. It is synthesized at the ribosomes. The preproinsulin polypeptide consists of four regions. These regions are P, B, C, and A.
- Regions A and B become A and B chains of insulin molecule.
 - C is the segment connecting these chains. It is cleaved away.
 - The P segment functions as a leader or signal sequence. It helps in the passage of polypeptide chain through membrane of ER into its lumen. It is enzymatically cleaved from the polypeptide.

Preproinsulin is produced and processed by a single type of cell, pancreatic cell. But other preprohormones are synthesized by different types of cells. They are processed into different end products for secretion. Its example is preproopiomelanocortin. It is the precursor of proopiomelanocortin (POMC). POMC is synthesized by different types of cells, including cells in the anterior pituitary gland, brain, skin, and mammalian placenta.

2. Formation Proinsulin

Polypeptide is called **proinsulin** without P segment. Proinsulin contains sulfhydryl (—SH) groups. These groups are oxidized to three disulfide (—S—S—) bridges (bonds) that fold the molecule. After the disulfide bridges form, C segment is removed by enzymatic action. It leaves A and B chains linked by disulfide bonds. These stages of posttranslational processing occur as the proinsulin molecules travel first to the Golgi apparatus in shuttle vesicles. Then it moves as secretory vesicles that bud off from the

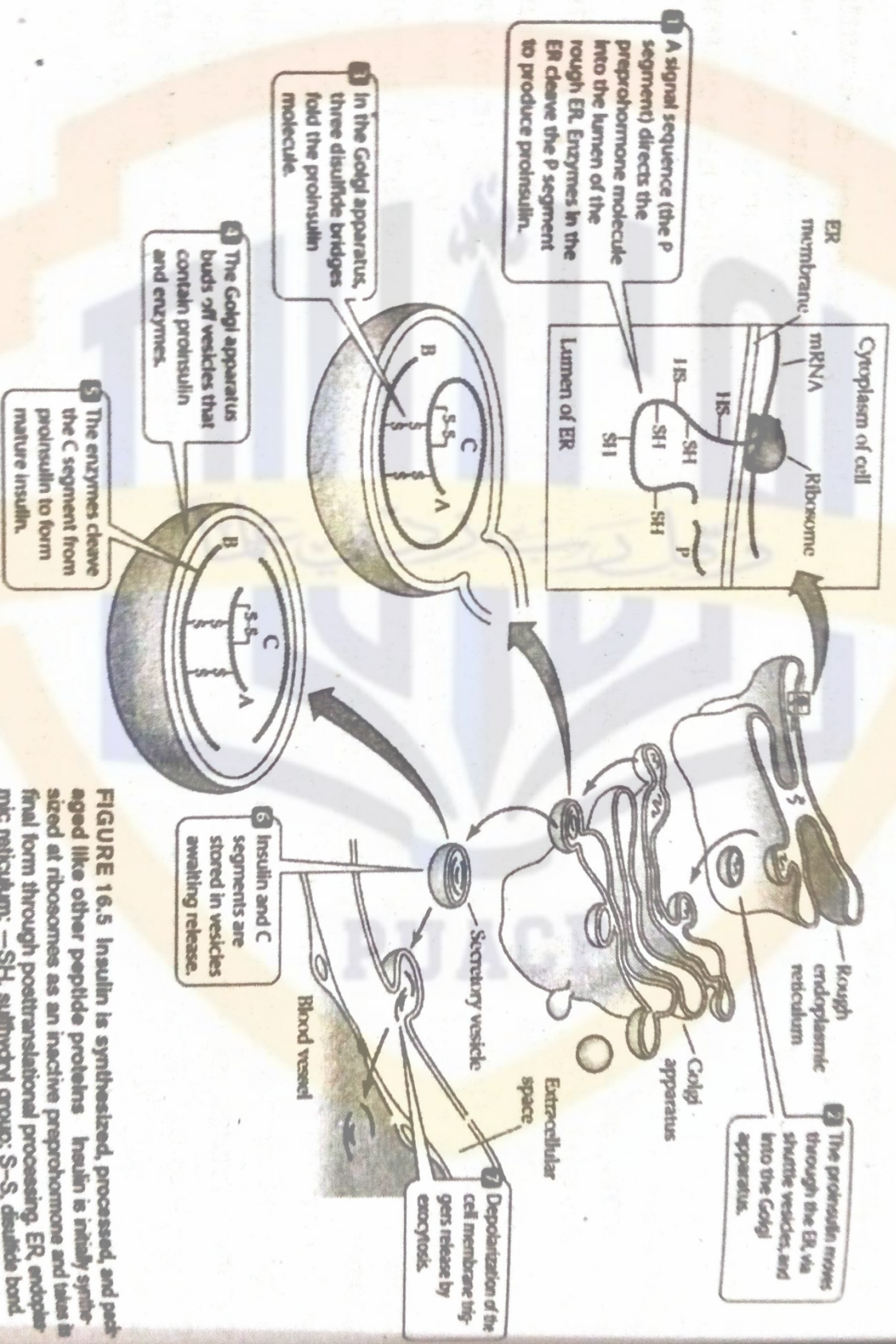


FIGURE 16.5 Insulin is synthesized, processed, and packaged like other peptide proteins. Insulin is initially synthesized at ribosomes as an inactive preprohormone and takes its final form through posttranslational processing. ER, endoplasmic reticulum; —SH, sulfhydryl group; S—S, disulfide bond.

Fig: Insulin is synthesized, processed, and packaged like other peptide proteins

Golgi apparatus. The C fragments are retained in the secretory vesicles. These are secreted along with the mature insulin molecules. The C fragment is not physiologically active. It is metabolized in the blood.

3. Storage in vesicles

Insulin and other secreted peptides are stored in vesicles within the cells. They are secreted by calcium-dependent exocytosis. When blood concentrations of glucose increase, cells depolarize. It causes voltage-gated Ca^{2+} channels to open and allow an influx of Ca^{2+} ions. The rate of insulin secretion is controlled by several factors. The most important factor is levels of glucose and certain amino acids in the blood. By this negative feedback mechanism, the blood glucose concentration is kept relatively constant.

Synthesis of Steroid hormones

Steroid hormones are synthesized on demand prior to secretion. These are released into the blood by diffusion. Steroid hormones are synthesized from cholesterol. In vertebrates, some cholesterol is obtained from animal fats in the diet. Steroidogenic (steroid-producing) endocrine cells and liver cells also synthesize cholesterol. Cholesterol is not water-soluble. Therefore, it circulates in the blood bound to lipoprotein carrier complexes. Low-density lipoprotein (LDL) complexes transport cholesterol to cells throughout the body. The high-density lipoprotein (HDL) complexes carry cholesterol away from cells to the liver.

1. **Obtaining cholesterol:** Steroidogenic cells express LDL receptor molecules on their cell membranes. An LDL complex binds to membrane receptor. The cell engulfs the complex by endocytosis. Thus it obtains cholesterol. In vertebrates, endocrine cells of the gonads, adrenal cortex, skin, and (in mammals) placenta produce steroid hormones.

2. **Synthesis of pregnenolone:** The cells of different endocrine glands synthesize or take in the cholesterol. Then the enzymes cleave the six-carbon side chain to form pregnenolone

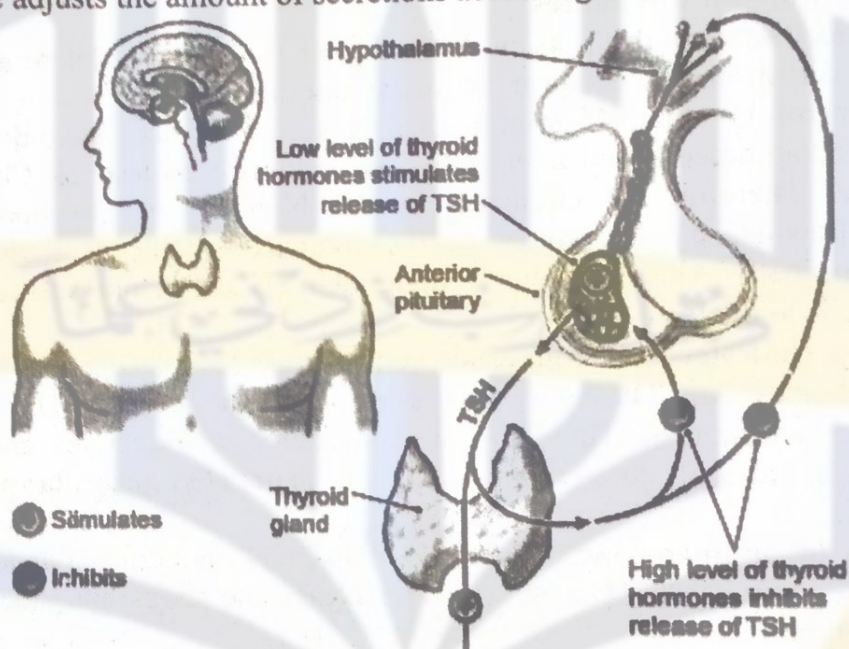
Different types of steroidogenic cells have different sets of enzymes. Hence each produces a different major end product. The enzymes necessary for steroid synthesis are contained within intracellular compartments. These enzymes involved in converting cholesterol to pregnenolone are located in the mitochondria. Most of the others are located in the smooth ER. Therefore, pregnenolone molecules must travel from mitochondria to smooth ER for further enzymatic conversions.

3. **Release from cells:** Steroid hormones are not stockpiled in vesicles prior to secretion. Instead, when the cell is stimulated, they are made from precursors stored in lipid droplets. Then it is immediately secreted. Secretion is accomplished by diffusion through the cell membrane. The blood concentration of a steroid hormone is determined by the rate of synthesis.

FEEDBACK MECHANISM

It is a type of interaction in which a controlling mechanism is itself controlled by the products of reactions it is controlling. Two opposing systems are needed for proper body functioning. If there are accelerators, there must be inhibitors. If one hormone in the body promotes or stimulates a reaction, another hormone is checking the same reaction. Interactions are mainly maintained in the body by feedback mechanism. In this way, concentrations of secretions are controlled automatically. There are two steps in feedback mechanism:

- Certain information about the activity of the body are passed or fed backed to the source.
- The source adjusts the amount of secretions according to this information.



Examples of feed back

The interaction between the pituitary and other endocrine glands is an example of feedback mechanism. This mechanism is very common in living systems. The feedback in thyroid gland function is described below.

Feedback in thyroid

- Low body temperature or stress stimulates neurosecretory cells of the hypothalamus. These secretory cells produce thyrotrophin releasing factor (TRF)
- TRF stimulate the anterior thyroid gland to produce Thyroid-stimulating hormone (TSH).
- The TSH then stimulates the thyroid gland to release thyroxine.
- Thyroxine increases the metabolic activity of most body cells and produce energy in the form of ATP and heat.
- Both ATP and heat raised the temperature body.
- Higher thyroxine levels in the blood inhibit the releasing-hormone cells and the TSH-